

Research Proposal - Complexity Costs of Fault-Tolerance Circuits.

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Abstract

We aim to study the overhead costs of fault-tolerant circuits in terms of both depth and width expansions. This means we ask what the optimal cost factor is, in each of the parameters alone, that one can hope to achieve, and also raise the question about the trade-off relation between the two. We ask these questions from both classical and quantum perspectives. The following proposal formulates the problems and states subproblems that might shed light on the answer.

We focus on shallow depth circuit classes, in particular, QAC_0 , $\text{QNC}_{0,f}$, QNC_1 , and 'QNC without reset gates', and certain known problem candidates for demonstrating quantum advantage such as factoring [Sho97] and Instantaneous Quantum Polynomial-time [BMS17], [Pal+24]. We aim to answer whether there exists a fault tolerance with only constant overhead at the circuit's depth. A positive answer implies that computing logarithmic depth circuits when subjected to noise is not harder than computing in an ideal environment.

1 Introduction.

The Quantum Computation model is widely believed to be superior to classical models, offering asymptotic speedup in tasks such as factoring [Sho97], searching [Gro96], simulating, and more relative to the best-known classical solutions. Yet even though there is almost complete agreement about the superiority of the ideal quantum model, there is still a debate over whether it is possible to implement complex computation in the real world, where the qubits and gates are subject to faults. Similarly, the feasibility of realizing classical computation has also been an open question. In fact the question about the feasibility of computation under noise is almost as ancient as the computer science field itself, initialized by Von Neumann [Neu56] at the time that classical computation putted in debuts. Time been pass and the followed works had pointed that not even a polynomial computation in the presence of noise is still reasonable but one can implement a fault tolerance version at a most constant times cost at the circuits depth [Pip85]. Or in asymptotic sense, classical computation in the presence of noise is as exactly hard as computation in ideal environment.

Recently, the feasibility question has been raised again, this time regarding quantum circuits, and while an intensive work has been done, and also succeed to prove that polynomial quantum computation can be made fault tolerance, [AB99],[Got14] and even with an only constant overhead at the original circuit width [Gro19], polylogarithmic depth [TKY24], and logarithmic depth [NP24]. Yet the minimal required depth overhead is still not well understood. That raises our first open problem:

Open problem 1.1. Is there a fault-tolerance scheme that maps circuits to noise-tolerant versions with a depth that is at most a constant times more? If not, what is the lower bound for depth overhead?

We stress that in all the mentioned constructions, the original constant-depth gates are mapped to logical presentations whose underlying implementation has asymptotically growing depth. This is in contrast to Pippenger [Pip85], where the logical gates are $O(1)$ -depth circuits.

Such an expansion prevents any construction that merely shifts presentations without interfering with the logical computation path of the original circuits from reaching a constant

depth overhead. Previous works have made way to bypass this obstacle; notably, [Ngu24] and [Li+25] present good error correction codes with transversal **CCZ** gates, and the series of works by Bombin [BM06], [Bom10], [Bom15] yield the 2D-color and 3D-color codes, which can be embedded on low-dimensional lattices (2D and 3D) and offer transversal gate implementation, where the first offers the whole Clifford group and the second the **T** gate. However, no code with a non-trivial distance and $O(1)$ -depth implementation for a complete universal gate set is known what raises our following open problem:

Open problem 1.2. Is there an error correction code with non-trivial distance for which a full universal gate set over the logical qubits can be realized at constant depth?

Since error correction codes with trivial distance might have a constant depth realization of a universal gate set (take, for example, the code where any codeword consists of a single physical qubit that encodes the logical state, adjacent to garbage), it is also interesting to ask whether there is a turning point:

Open problem 1.3. If the answer to Open problem 1.2 is yes, what is the minimum distance for which there exists a code with that distance, and the code provides a universal gate set realization at constant depth?

Similarly, one can also ask whether the code dimension plays a role in the feasibility of realizing logical gates. Recent work provides a first result regarding the width-depth tradeoff. Specifically, they show that for classical codes with distance d , for any logical qubit i , there exists a logical qubit $j \neq i$ such that the code offers the $\mathbf{CX}_{i,j}$ logical gate at most at l -depth. Then, it must hold that $k = O\left(\frac{n}{d^{2-l}}\right)$. They also mention that their proof does not imply an analogous tradeoff relation for quantum codes. This brings us to ask if the result can be generalized to the implementation of general gates and to quantum error correction codes.

Open problem 1.4. Whether there is a relation between the depth of a logical gate of an error correction code and the dimension and depth of that code. Is that relation identical for both classical and quantum codes? If not, how does the trade-off behave in each case?

Zooming out from the operations over code blocks raises the question in the context of fault-tolerant circuits. In classical settings, lower bounds on the overall circuit size were established. A series of works [Gal91], [RS91], showed that fault-tolerant circuits that compute Boolean functions that are s -sensitive, meaning that a change in one of s bits in the input leads to a different output, must exhibit at least $\Omega(s \log s)$ size. In particular, a fault-tolerant circuit that computes the n -bit **Xor** function must have $\Omega(n \log n)$ size.

Furthermore, [Rom06] exhibits a trade-off in a model in which the given input is already encoded. In that model, they provide a fault-tolerant circuit with a width overhead of the circuit being $\Theta\left(\frac{\log n}{\log \log n}\right)$, while the depth overhead is $\Theta(\log \log n)$. So, it's attempting to ask if for any w and d such that $w \cdot d = \log(n)$, a fault-tolerant circuit with w and d width and depth overhead can be realized:

Open problem 1.5. Consider a circuit over n classical (quantum) bits. What are the domains of depth d and width w for which the circuit can be fault-tolerant at the cost of w and d width and depth overheads? How does the answer change if we assume encoded input?

The questions above address the fault tolerance of general circuits, but focusing on shallow circuits also attracts special interest, particularly for near-term applications. Since today's quantum machines struggle to maintain quantum states for extended periods, researchers were motivated to define **NISQ**, which stands for Noisy Intermediate-Scale Quantum, referring to the current era of quantum computing characterized by quantum processors with a limited number of qubits and susceptibility to errors due to noise. Nevertheless, there are certain known problem candidates for demonstrating quantum advantage, such as factoring [Sho97] and Instantaneous Quantum Polynomial-time [BMS17], [Pal+24], which have been proven to be solvable by sampling from logarithmic-depth quantum circuits. We emphasize that in [Pal+24] they showed a realization that uses constant quantum depth circuits with the intermediation of a polynomial classical circuit in the middle. From a technical perspective, that means that to implement the protocol, one has to protect the quantum state for the time that the classical routine lasts. Hence, it is interesting to ask what both quantum and classical low-resource fault-tolerant realizations can be designed for those particular problems.

Open problem 1.6. Does sampling from **IQP** have a fault-tolerant realization using $\log n$ depth circuits, namely an $O(1)$ -depth overhead? Or at least, is it possible to postpone all the

classical computation until after the qubits are measured? Are there any other candidates for demonstrating quantum advantage that can be realized using shallow circuits?

Another application that uses logarithmic depth is encoding. It was shown in [MN98] that stabilizer codes can be prepared at logarithmic depth, and not only that, but the preparation circuit is composed of only Hadamard and controlled-Pauli gates. In other words, the family of encoder circuits is much less rich than general circuits, and therefore, it is interesting to ask about the synthesis of encoded codewords. By synthesis, we mean generating the encoded codeword itself and not a logical representation of it (not a concatenation with another code).

Open problem 1.7. Do encoders have a fault-tolerant realization with an $O(1)$ -depth overhead?

In addition to NISQ, another common characterization for limited quantum computation is computation without reset gates — namely, the gates which initialize qubits to the zero state $|0\rangle$. It was proved that in this model, when restricted to polynomial space, any computation that lasts longer than logarithmic depth results in noise [Aha+96]. Having a constant depth-overhead fault tolerance scheme would imply the feasibility of log depth computation in that model.

Open problem 1.8. Assuming there are $\Theta(\log(n))$ -depth circuits that can be fault-tolerized at the cost of $O(1)$ -depth overhead, could those constructions be generalized to hold in the non-reset gates regime?

Last but not least, we would like to connect fault-tolerance results to complexity theory in a broad manner. Historically, new inventions of error correction codes have stimulated advances in PCP theory and vice versa. For instance, the left-right Cayley complex [Din+22], [LZ22], [PK21] was used recently in [BMV24] to design a routing net that is robust to collapses, which in turn was used to construct a proof-checking scheme in a manner similar to the Polishchuk-Spielman approach [PS94]. Another recent result showed that having an $O(1)$ -depth overhead fault tolerance protocol implies that quantum PCP with logarithmic soundness could be achieved [ABN24]. Thus, we suggest the following vague question:

Open problem 1.9. Whether fault-tolerant construction can help to design a verification scheme? Does an analog to the robustness routing protocol exist for verifying quantum computations? What else can be borrowed from the fault-tolerant world for the use of quantum PCP?

References

- [Neu56] J. von Neumann. “Probabilistic Logics and the Synthesis of Reliable Organisms From Unreliable Components”. In: *Automata Studies*. Ed. by C. E. Shannon and J. McCarthy. Princeton: Princeton University Press, 1956, pp. 43–98. ISBN: 9781400882618. DOI: [doi:10.1515/9781400882618-003](https://doi.org/10.1515/9781400882618-003). URL: <https://doi.org/10.1515/9781400882618-003>.
- [Pip85] Nicholas Pippenger. “On networks of noisy gates”. In: *26th Annual Symposium on Foundations of Computer Science (sfcs 1985)*. 1985, pp. 30–38. DOI: [10.1109/SFCS.1985.41](https://doi.org/10.1109/SFCS.1985.41).
- [Gal91] A. Gal. “Lower bounds for the complexity of reliable Boolean circuits with noisy gates”. In: *[1991] Proceedings 32nd Annual Symposium of Foundations of Computer Science*. 1991, pp. 594–601. DOI: [10.1109/SFCS.1991.185424](https://doi.org/10.1109/SFCS.1991.185424).
- [RS91] R. Reischuk and B. Schmolz. “Reliable computation with noisy circuits and decision trees—a general $n \log n$ lower bound”. In: *[1991] Proceedings 32nd Annual Symposium of Foundations of Computer Science*. 1991, pp. 602–611. DOI: [10.1109/SFCS.1991.185425](https://doi.org/10.1109/SFCS.1991.185425).
- [PS94] Alexander Polishchuk and Daniel A. Spielman. “Nearly-linear size holographic proofs”. In: *Proceedings of the Twenty-Sixth Annual ACM Symposium on Theory of Computing*. STOC ’94. Montreal, Quebec, Canada: Association for Computing Machinery, 1994, pp. 194–203. ISBN: 0897916638. DOI: [10.1145/195058.195132](https://doi.org/10.1145/195058.195132). URL: <https://doi.org/10.1145/195058.195132>.

- [Aha+96] D. Aharonov et al. *Limitations of Noisy Reversible Computation*. 1996. arXiv: [quant-ph/9611028](https://arxiv.org/abs/quant-ph/9611028) [quant-ph]. URL: <https://arxiv.org/abs/quant-ph/9611028>.
- [Gro96] Lov K. Grover. *A fast quantum mechanical algorithm for database search*. 1996. arXiv: [quant-ph/9605043](https://arxiv.org/abs/quant-ph/9605043) [quant-ph].
- [Sho97] Peter W. Shor. “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer”. In: *SIAM Journal on Computing* 26.5 (Oct. 1997), pp. 1484–1509. DOI: [10.1137/s0097539795293172](https://doi.org/10.1137/s0097539795293172). URL: <https://doi.org/10.1137%2Fs0097539795293172>.
- [MN98] Cristopher Moore and Martin Nilsson. *Parallel Quantum Computation and Quantum Codes*. 1998. arXiv: [quant-ph/9808027](https://arxiv.org/abs/quant-ph/9808027) [quant-ph].
- [AB99] Dorit Aharonov and Michael Ben-Or. *Fault-Tolerant Quantum Computation With Constant Error Rate*. 1999. arXiv: [quant-ph/9906129](https://arxiv.org/abs/quant-ph/9906129) [quant-ph].
- [BM06] H. Bombin and M. A. Martin-Delgado. “Topological Quantum Distillation”. In: *Phys. Rev. Lett.* 97 (18 Oct. 2006), p. 180501. DOI: [10.1103/PhysRevLett.97.180501](https://doi.org/10.1103/PhysRevLett.97.180501). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.97.180501>.
- [Rom06] Andrei Romashchenko. “Reliable computations based on locally decodable codes”. In: *Proceedings of the 23rd Annual Conference on Theoretical Aspects of Computer Science*. STACS’06. Marseille, France: Springer-Verlag, 2006, pp. 537–548. ISBN: 3540323015. DOI: [10.1007/11672142_44](https://doi.org/10.1007/11672142_44). URL: https://doi.org/10.1007/11672142_44.
- [Bom10] H. Bombin. “Topological subsystem codes”. In: *Phys. Rev. A* 81 (3 Mar. 2010), p. 032301. DOI: [10.1103/PhysRevA.81.032301](https://doi.org/10.1103/PhysRevA.81.032301). URL: <https://link.aps.org/doi/10.1103/PhysRevA.81.032301>.
- [Got14] Daniel Gottesman. *Fault-Tolerant Quantum Computation with Constant Overhead*. 2014. arXiv: [1310.2984](https://arxiv.org/abs/1310.2984) [quant-ph].
- [Bom15] Héctor Bombín. “Gauge color codes: optimal transversal gates and gauge fixing in topological stabilizer codes”. In: *New Journal of Physics* 17.8 (Aug. 2015), p. 083002. DOI: [10.1088/1367-2630/17/8/083002](https://doi.org/10.1088/1367-2630/17/8/083002). URL: <https://doi.org/10.1088/1367-2630/17/8/083002>.
- [BMS17] Michael J. Bremner, Ashley Montanaro, and Dan J. Shepherd. “Achieving quantum supremacy with sparse and noisy commuting quantum computations”. In: *Quantum* 1 (Apr. 2017), p. 8. ISSN: 2521-327X. DOI: [10.22331/q-2017-04-25-8](https://doi.org/10.22331/q-2017-04-25-8). URL: <https://doi.org/10.22331/q-2017-04-25-8>.
- [Gro19] Antoine Grospellier. “Constant time decoding of quantum expander codes and application to fault-tolerant quantum computation”. Theses. Sorbonne Université, Nov. 2019. URL: <https://theses.hal.science/tel-03364419>.
- [PK21] Pavel Panteleev and Gleb Kalachev. *Asymptotically Good Quantum and Locally Testable Classical LDPC Codes*. 2021. DOI: [10.48550/ARXIV.2111.03654](https://doi.org/10.48550/ARXIV.2111.03654). URL: <https://arxiv.org/abs/2111.03654>.
- [Din+22] Irit Dinur et al. *Good Locally Testable Codes*. 2022. DOI: [10.48550/ARXIV.2207.11929](https://doi.org/10.48550/ARXIV.2207.11929). URL: <https://arxiv.org/abs/2207.11929>.
- [LZ22] Anthony Leverrier and Gilles Zémor. *Quantum Tanner codes*. 2022. arXiv: [2202.13641](https://arxiv.org/abs/2202.13641) [quant-ph].
- [ABN24] Anurag Anshu, Nikolas P. Breuckmann, and Quynh T. Nguyen. “Circuit-to-Hamiltonian from Tensor Networks and Fault Tolerance”. In: *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*. STOC ’24. ACM, June 2024, pp. 585–595. DOI: [10.1145/3618260.3649690](https://doi.org/10.1145/3618260.3649690). URL: <http://dx.doi.org/10.1145/3618260.3649690>.

- [BMV24] Mitali Bafna, Dor Minzer, and Nikhil Vyas. *Quasi-Linear Size PCPs with Small Soundness from HDX*. 2024. arXiv: [2407.12762](https://arxiv.org/abs/2407.12762) [cs.CC]. URL: <https://arxiv.org/abs/2407.12762>.
- [Ngu24] Quynh T. Nguyen. *Good binary quantum codes with transversal CCZ gate*. 2024. arXiv: [2408.10140](https://arxiv.org/abs/2408.10140) [quant-ph]. URL: <https://arxiv.org/abs/2408.10140>.
- [NP24] Quynh T. Nguyen and Christopher A. Pattison. “Quantum Fault Tolerance with Constant-Space and Logarithmic-Time Overheads”. In: *Proceedings of the 57th Annual ACM Symposium on Theory of Computing* (2024). URL: <https://api.semanticscholar.org/CorpusID:273850180>.
- [Pal+24] Louis Paletta et al. “Robust sparse IQP sampling in constant depth”. In: *Quantum* 8 (May 2024), p. 1337. ISSN: 2521-327X. DOI: [10.22331/q-2024-05-06-1337](https://doi.org/10.22331/q-2024-05-06-1337). URL: <https://doi.org/10.22331/q-2024-05-06-1337>.
- [TKY24] Shiro Tamiya, Masato Koashi, and Hayata Yamasaki. *Polylog-time- and constant-space-overhead fault-tolerant quantum computation with quantum low-density parity-check codes*. 2024. arXiv: [2411.03683](https://arxiv.org/abs/2411.03683) [quant-ph]. URL: <https://arxiv.org/abs/2411.03683>.
- [Li+25] Yiming Li et al. *Poincaré Duality and Multiplicative Structures on Quantum Codes*. 2025. arXiv: [2512.21922](https://arxiv.org/abs/2512.21922) [quant-ph]. URL: <https://arxiv.org/abs/2512.21922>.